

Mathematics, 2017 EC, entrance exam, version 4

Grade 12 · Mathematics · 2017 E.C.

Time allowed: 120 minutes

Questions: 59

Total marks: 59

Name _____ Grade _____ ID No. _____ Date _____

1. After rationalizing the denominator of $5/(3\sqrt{3} - 2\sqrt{2})$, the denominator is equal to _____. (1)
 A. 35 B. $5(3\sqrt{3}-2\sqrt{2})/19$
 C. 19 D. $5(3\sqrt{3}+2\sqrt{2})/19$
2. The solution set of the equation $5|2x-1| - 3|4x+5| = 2$ is equal to: (1)
 A. $\{6/11\}$ B. $\{-11, -9, -6/11\}$
 C. $\{\}$ D. $\{-9, -6/11\}$
3. In a survey of 600 students, 250 drink tea, 225 drink coffee, 75 drink both. How many drink neither? (1)
 A. 400 B. 550
 C. 200 D. 125
4. Given relation $R = \{(x,y): y > x-2, y > -x-2, y \leq 4\}$. Which is the domain and range of R respectively? (1)
 A. $(-6,6)$ and $(-2,4]$ B. $(-2,4]$ and $(-6,6)$
 C. $[-6,6]$ and $(-2,4)$ D. $(-2,4)$ and $(-6,6)$
5. $\triangle ABC$ and $\triangle DEF$ are similar. Area of $\triangle ABC = 12 \text{ m}^2$. DE is twice AB. Area of $\triangle DEF$? (1)
 A. 6 m^2 B. 48 m^2
 C. 36 m^2 D. 24 m^2
6. Vector AB has initial A(2,-1) and terminal B(-4,7). Which is a unit vector in same direction? (1)
 A. $-(3/5)i + (4/5)j$ B. $(3/5)i - (4/5)j$
 C. $-(3/50)i - (4/50)j$ D. $(3/50)i + (4/50)j$
7. Range of $f(x) = (x+2)/(x-2)$? (1)
 A. $\mathbb{R} \setminus \{2\}$ B. $\mathbb{R} \setminus \{-2\}$
 C. $\mathbb{R} \setminus \{-1\}$ D. $\mathbb{R} \setminus \{1\}$
8. Firm A offers starting salary 5000 Birr with raise of 600 per year. Total earned over first 20 years? (1)
 A. 214,000 B. 16,400
 C. 264,000 D. 216,000
9. Arithmetic sequence: first term 10, tenth term -26. Find a_{101} . (1)
 A. 420 B. 380
 C. -410 D. -390
10. Area enclosed by $f(x)=1-x^2$ and x-axis from $x=0$ to $x=2$. (1)
 A. 3 B. 2
 C. $2/3$ D. $4/3$

11. Constant term in expansion of $(x - 2/x^2)^6$. (1)

A. -160
B. -60
C. 60
D. 240

12. Square ABCD inscribed in circle radius 4 cm. Area of segment cut off by chords AB, BC, CD, AD? (1)

A. $16(\pi+2)$
B. $16(\pi-2)$
C. $16(\pi+1)$
D. $16(\pi+1)$

13. Which statement is NOT true? (1)

A. $x+2$ is factor of $x^3-3x^2-4x+12$
B. $f(x)=x^4-6x^3+x^2+12x-6$ has real root between -1 and 0
C. Dividing $x^4+6x^3-10x+3$ by x^2-1 gives remainder $-4x+4$
D. Graph of degree n has at most $n-1$ turning points

14. P(-2,5) and Q(8,-1). Q is midpoint of PR. Find R. (1)

A. (-7,18)
B. (14,3)
C. (18,-7)
D. (3,2)

15. Mr. Soresa divided 40,000 for three workers in ratio $1/2 : 3/10 : 1/5$. Find C's share. (1)

A. 20,000
B. 8,000
C. 12,000
D. 13,333.33

16. Solution set of $\log_2(x+2)+\log_2(x-1)=2$. (1)

A. $\{-3\}$
B. $\{-3,2\}$
C. $\{\}$
D. $\{2\}$

17. If $\cos \theta = -3/5$ and $\tan \theta < 0$, which is true? (1)

A. $\sin \theta = -4/5$
B. $\cot \theta = -4/3$
C. $\csc \theta = 5/4$
D. $\tan \theta = 4/3$

18. Volume of frustum of pyramid with base areas 81 cm^2 and 121 cm^2 , altitude 6 cm. (1)

A. 602π
B. 602
C. 301
D. 903π

19. Simplify $(x^2-4)/(x^2+3x-10) \div (x^2+5x+6)/(x^2+8x+15)$. (1)

A. $(x+2)/(x+5)$
B. $((x+2)/(x+5))^2$
C. $(x-2)/(x+3)$
D. none

20. If $f(x) = (\pi^2+1)^4$, then $f'(x) = ?$ (1)

A. 0
B. $8\pi(\pi^2+1)^3$
C. $4(\pi^2+1)^3$
D. $8(\pi^2+1)^3$

21. Which is a geometric sequence? (1)

A. 1, 1.5, 2.25, 3.375, ...
B. 1, 1/8, 1/32, 1/64, ...
C. 27, 21, 15, 9, ...
D. 1, 4, 8, 12, 15, ...

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- 22.** For matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & -5 \\ 3 & -5 & 6 \end{bmatrix}$, which is NOT true? (1)
- A. A is symmetric
B. $\det(A) = -71$
C. A is invertible
D. Cofactor of $a_{32} = 11$
-
- 23.** Equation of tangent to circle $x^2 + y^2 - 8x - 10y - 128 = 0$ at $(-8, 10)$. (1)
- A. $5x - 12y + 136 = 0$
B. $12x - 5y + 146 = 0$
C. $12x + 5y + 146 = 0$
D. $5x + 12y + 136 = 0$
-
- 24.** Image of circle $x^2 + y^2 - 4x + 6y + 12 = 0$ rotated by $\pi/2$ about origin. (1)
- A. $(x-3)^2 + (y-2)^2 = 1$
B. $(x+3)^2 + (y+2)^2 = 0$
C. $(x-3)^2 + (y+2)^2 = 1$
D. $(x+3)^2 + (y-2)^2 = 1$
-
- 25.** Which vector is orthogonal to $u = (-4, 6)$? (1)
- A. $(-3, 2)$
B. $(-1, -1)$
C. $(3, -2)$
D. $(-3, -2)$
-
- 26.** Which is NOT true about transformation? (1)
- A. Reflection is onto mapping
B. Translation moves rectangles to congruent rectangles
C. Rotation moves circles to congruent circles
D. Reflection is non-rigid motion
-
- 27.** Balloon inflated at $900 \text{ cm}^3/\text{s}$. Rate of radius increase when radius = 15 cm. (1)
- A. $\pi \text{ cm/s}$
B. 1 cm/s
C. $1/\pi \text{ cm/s}$
D. $1/4 \text{ cm/s}$
-
- 28.** Marginal revenue when $x=20$ for $R(x) = 3x^2 + 36x + 5$. (1)
- A. 156
B. 120
C. 1925
D. 1236
-
- 29.** Vectors $a = 2i - 3j + 2k$, $b = 2i + 3j + k$. Area of triangle OAB. (1)
- A. $229/2$
B. $\sqrt{85}$
C. $\sqrt{229}/2$
D. $2\sqrt{229}$
-
- 30.** Vector with initial $(2, 5, 0)$ and terminal $(-3, 7, 4)$. (1)
- A. $-5i + 2j - 4k$
B. $-5i + 2j + 4k$
C. $5i - 2j - 4k$
D. $-i + 12j + 4k$
-
- 31.** GCF of 65 and 117 expressed as $65m - 117$. Find m. (1)
- A. 1
B. 2
C. 3
D. 4
-
- 32.** Greatest ratio among 3:2, 8:5, 121:75, 5:3. (1)
- A. 5:3
B. 8:5
C. 121:75
D. 3:2

- 33.** Benti earns 18,000, Derartu 20,000. Ratio of Derartu's earnings to total? (1)
- A. 9:10
B. 10:19
C. 9:19
D. 10:18
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- 34.** Maximize $Z=4x+y$ subject to $x+y \leq 50$, $3x+y \leq 90$, $x,y \geq 0$. (1)
- A. 120
B. 110
C. 200
D. 50
-
- 35.** Linear programming constraints for screws A and B: 4 min automatic, 6 min hand for A; 6 min automatic, 3 min hand for B; each machine available 4 hours (240 min). (1)
- A. $4x+6y \leq 240$, $6x+3y \leq 240$, $x \geq 0, y \geq 0$
B. $4x+6y \geq 240$, $6x+3y \leq 240$
C. $4x+6y \geq 240$, $6x+3y \geq 240$
D. $3x+6y \geq 240$, ...
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- 36.** Corner points (0,0),(5,0),(3,4),(0,5). $Z=px+qy$, $p,q > 0$. Max occurs at (3,4) and (0,5). Find relation. (1)
- A. $q=3p$
B. $p=3q$
C. $p=q$
D. $p > (1/3)q$
-
- 37.** What percent of 15 is equal to 600? (1)
- A. 400
B. 90
C. 690
D. 4000
-
- 38.** Shop sells 30 articles for 6600. Plans to sell each for 200. Approximate discount percent? (1)
- A. 9.091%
B. 10%
C. 90.091%
D. 8%
-
- 39.** 12% tax on shoes amounts to 45 Birr. What is 8% tax? (1)
- A. 30
B. 37.5
C. 40
D. 35
-
- 40.** Corner points (0,2),(3,0),(6,0),(6,8),(0,5). Minimize $Z=4x+6y$. Minimum occurs at: (1)
- A. (0,2) only
B. (3,0) only
C. any point on line segment joining (0,2) and (3,0)
D. midpoint of that segment only
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- 41.** Corner points (0,0),(0,40),(20,40),(60,20),(60,0). Objective $Z=5x+6y$. Compare max Z (Column A) with 421 (Column B). (1)
- A. Column A greater
B. Column B greater
C. Equal
D. Cannot determine
-
- 42.** Box of 15 oranges, 12 good, 3 bad. Select 3 without replacement. Probability box approved (all three good). (1)
- A. 12/15
B. 1/5
C. 44/91
D. 88/225
-
- 43.** Profit 20% on selling price of 960 Birr. Find cost. (1)
- A. 600
B. 800
C. 900
D. 840

44. Infected 230,000, died 10,000, remaining 43% cured. Number still infected as percent of total? (1)

- A. 57% B. 54.5%
- C. 56.5% D. 65%

45. Meti sold shoe for 10,000 with 60% markup on selling price. Find cost price. (1)

- A. 4,000
B. 6,000
C. 16,000
D. 5,000

46. $A = \begin{bmatrix} -2 & 3 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$, $|B| = 1/3$. Find $\det(2A^T B)$. (1)

- A. 16
C. -16
B. -4
D. 48

47. A, B non-zero square matrices same order, $(A+B)^2 = A^2+B^2$. Then which is true? (1)

- A. $AB=0$
B. $BA=0$
C. $AB=BA$
D. $AB=-BA$

48. $P=(1,a,a)$, $Q=(a-1,1,1)$. Distance $PQ=3$. Find set containing a . (1)

- A. $\{-9,1\}$ B. $\{-1/3,3\}$
C. $\{-3,1/3\}$ D. $\{-1,9\}$

49. Probability of occurrence = $2/3$. Probability of non-occurrence? (1)

- A. $\frac{4}{3}$
C. $\frac{3}{4}$
- B. $\frac{2}{3}$
D. $\frac{1}{3}$

50. Red ball and 4 white balls. Draw two without replacement. Probability both white? (1)

- A. $\frac{1}{5}$
B. $\frac{4}{5}$
C. $\frac{16}{25}$
D. $\frac{3}{5}$

51. Which is necessarily true for vectors A and B? (1)

- A. If $|A|=|B|$ then $A=B$
- B. $|kA| = k|A|$ for $k \in \mathbb{R}$
- C. If u is unit vector in direction of A , then $A \cdot u = |A|$
- D. If A parallel to B , then $A \cdot B = 0$

52. Frequency table: $x:3,4,5,6,7$; $f:2,4,8,4,2$. Which is NOT true? (1)

- A. Mode = 5
B. Middle quartile = 5
C. Standard deviation = $\sqrt{1.2}$
D. 75th percentile = 6.5

53. Length 10m, breadth 2.4cm. Ratio length:breadth? (1)

- A. 5:1.2 B. 25:6
C. 1250:3 D. 625:6

54. Farmer bought bull 44,000, cow 18,000. Sold bull at 5% loss, cow at 10% profit. Net result? (1)

- A. loss 200
B. profit 400
C. loss 400
D. profit 200

55. Solution set of system: $x+y-z=1$, $x+2y-3z=1$, $2x+3y-4z=2$. (1)

A. $\{(1,1,0)\}$

B. $\{(1-k, 2k, k): k \in \mathbb{R}\}$

C. $\{\}$

D. $\{(2k, 1-k, k): k \in \mathbb{R}\}$

56. Which pair of lines are perpendicular? (1)

A. $y=2x-1$ and $y=2x+1$

B. $y+2x=1$ and $2y-x=-1$

C. $y-2x=-2$ and $y - (1/2)x=1$

D. $y=3x+1$ and $y=(1/3)x+1$

57. Weight data: 60,67,63,64,61,49,52,68,55,61. Which is NOT true about mean deviation? (1)

A. Mean deviation about mean = 4.8

B. Mean deviation about median = 4.6

C. Mean deviation about mode = 4.6

D. Mean deviation about mean = 4.6

58. Tent: cylinder height 5m, diameter 6m, conical top slant height 8m. Area of canvas? (1)

A. 54π

B. 39π

C. 78π

D. 63π

59. In circle, chords AC and BD intersect at P. $\angle DAC=35^\circ$, $\angle APB=65^\circ$. Find $\angle ACB$. (1)

A. 130°

B. 60°

C. 70°

D. 30°

— END OF EXAMINATION —